THE SCHEME THEORETIC UNION OF TWO CLOSED SUBSCHEMES IS THE PUSHOUT OF ONE ALONG THEIR SCHEME THEORETIC INTERSECTION, AND ITS STRUCTURE SHEAF SITS IN A SHORT EXACT SEQUENCE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The canonical Stacks tag of the counted statement is 0C4J (morphisms-lemma-scheme-theoretic-union). Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the tag is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The author's slip at proof line 610, where the restriction of g is written g|_{Y cap V} while the surrounding argument uses Y cap U, is reproduced verbatim; no mathematical text was corrected.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of sources/stacks-chow/preamble.tex lines 14-18 and 154-217. No macro was invented or redefined.

Editorial note. The definition of the scheme theoretic intersection and union is included as a context span because the statement names X union Y, X cap Y without repeating the definition.

Allowed standard prerequisites (editorial).

- Closed subschemes of a scheme correspond to quasi-coherent sheaves of ideals, and inclusion of subschemes is inclusion of ideals. Same chapter, Lemma labelled lemma-closed-immersion-ideals (canonical Stacks tag 01QP); invoked at proof line 577. Supplies that X -> X union Y and Y -> X union Y are closed immersions, and that the union is cut out by the intersection of the two ideal sheaves.
- For a closed immersion i : Z -> X with ideal sheaf I, the functor i_* identifies quasi-coherent O_Z-modules with quasi-coherent O_X-modules annihilated by I. Same chapter, Lemma labelled lemma-i-star-equivalence (canonical Stacks tag 01QY); invoked at proof line 578. Lets the author regard all four structure sheaves of the displayed sequence as quasi-coherent O_S-modules, so that exactness may be checked on stalks or affine locally.
- Morphisms of schemes may be constructed by glueing local data. Schemes, Section labelled schemes-section-glueing-schemes (canonical Stacks tag 01JA); invoked at proof line 601. Justifies building the morphism h : X union Y -> T from its restrictions to the three regions of the proof.

Source reference. Original lemma 01QP: sources/stacks-morphisms/pinned-original-source.tex, source lines 110–110. Same-chapter pointer invoked at proof line 577

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for the ideal-sheaf description of closed subschemes; the ideal sheaves I and J themselves are named in the counted definition and the conclusion that the union is cut out by I cap J is used inside the counted proof.

Source reference. Original lemma 01QY: sources/stacks-morphisms/pinned-original-source.tex, source lines 386–386. Same-chapter pointer invoked at proof line 578 as a way of viewing the four structure sheaves as quasi-coherent O_S-modules; all exactness and glueing arguments that use this point of view are carried out inside the counted proof.

Source reference. Original section 01JA: sources/stacks-chow/tags.txt, source lines 2007–2007. Cross-chapter pointer invoked at proof line 601 for the licence to construct a morphism from local data; the local construction of h and the case analysis are carried out in full inside the counted span.

Lemma 0.1. *Let S be a scheme. Let $X, Y \subset S$ be closed subschemes. Let $X \cup Y$ be the scheme theoretic union of X and Y. Let $X \cap Y$ be the scheme theoretic intersection of X and Y. Then $X \to X \cup Y$ and $Y \to X \cup Y$ are closed immersions, there is a short exact sequence*

$$0 \to \mathcal{O}_{X \cup Y} \to \mathcal{O}_X \times \mathcal{O}_Y \to \mathcal{O}_{X \cap Y} \to 0$$

of $\mathcal{O}_S$-modules, and the diagram

$$\begin{array}{ccc} X \cap Y & \longrightarrow & X \\ \downarrow & & \downarrow \\ Y & \longrightarrow & X \cup Y \end{array}$$

is cocartesian in the category of schemes, i.e., $X \cup Y = X \amalg_{X \cap Y} Y$.

Proof. The morphisms $X \to X \cup Y$ and $Y \to X \cup Y$ are closed immersions by Lemma 01QP. In the short exact sequence we use the equivalence of Lemma 01QY to think of quasi-coherent modules on closed subschemes of S as quasi-coherent modules on S. For the first map in the sequence we use the canonical maps $\mathcal{O}_{X \cup Y} \to \mathcal{O}_X$ and $\mathcal{O}_{X \cup Y} \to \mathcal{O}_Y$ and for the second map we use the canonical map $\mathcal{O}_X \to \mathcal{O}_{X \cap Y}$ and the negative of the canonical map $\mathcal{O}_Y \to \mathcal{O}_{X \cap Y}$. Then to check exactness we may work affine locally. Let $U = \mathrm{Spec}(A)$ be an affine open of S and let $X \cap U$ and $Y \cap U$ correspond to the ideals $I \subset A$ and $J \subset A$. Then $(X \cup Y) \cap U$ corresponds to $I \cap J \subset A$ and $X \cap Y \cap U$ corresponds to $I + J \subset A$. Thus exactness follows from the exactness of

$$0 \to A/I \cap J \to A/I \times A/J \to A/(I + J) \to 0$$

To show the diagram is cocartesian, suppose we are given a scheme T and morphisms of schemes $f : X \to T$, $g : Y \to T$ agreeing as morphisms $X \cap Y \to T$. Goal: Show there exists a unique morphism $h : X \cup Y \to T$ agreeing with f and g. To construct h we may work affine locally on $X \cup Y$, see Schemes, Section 01JA. If $s \in X$, $s \notin Y$, then $X \to X \cup Y$ is an isomorphism in a neighbourhood of s and it is clear how to construct h. Similarly for $s \in Y$, $s \notin X$. For $s \in X \cap Y$ we can pick an affine open $V = \mathrm{Spec}(B) \subset T$ containing $f(s) = g(s)$. Then we can choose an affine open $U = \mathrm{Spec}(A) \subset S$ containing s such that $f(X \cap U)$ and $g(Y \cap U)$ are contained in V. The morphisms $f|_{X \cap U}$ and $g|_{Y \cap V}$ into V correspond to ring maps

$$B \to A/I \quad \text{and} \quad B \to A/J$$

which agree as maps into $A/(I + J)$. By the short exact sequence displayed above there is a unique lift of these ring homomorphism to a ring map $B \to A/I \cap J$ as desired. □

Definition 0.2. Let X be a scheme. Let $Z, Y \subset X$ be closed subschemes corresponding to quasi-coherent ideal sheaves $\mathcal{I}, \mathcal{J} \subset \mathcal{O}_X$. The *scheme theoretic intersection* of Z and Y is the closed subscheme of X cut out by $\mathcal{I} + \mathcal{J}$. The *scheme theoretic union* of Z and Y is the closed subscheme of X cut out by $\mathcal{I} \cap \mathcal{J}$.